

3. Learning algorithms. Inspired by Hebbian learning principles, allow the system to adapt and learn from data

The role of memristors in neuromorphic computing

Memristors are particularly well-suited for neuromorphic computing

due to their synapse-like behaviour, enabling efficient implementation of artificial neural networks.

Advantages. They offer several advantages, such as the ability to store analogue values, which is beneficial for representing synaptic weights in neural networks. Additionally, memristors can change resistance states in response to electrical stimuli, mimicking the adaptive behaviour of biological synapses. Their nanoscale size allows for the creation of dense neural networks, enhancing computational power without a proportional increase in physical size or power consumption.

Applications. In practical applications, memristors can be used to implement the synaptic weights in artificial

neural networks (ANNs), providing a more efficient and scalable solution compared to traditional CMOS technology. They are also ideal for spiking neural networks (SNNs), where information is transmitted as discrete spikes, similar to the brain's communication method. The non-linear dynamics of memristors make them suitable for processing and storing spike-timing information. Furthermore, by integrating memory and computation, memristor-based systems can perform data-intensive tasks directly within the memory, reducing latency and power consumption.

Memristors represent a transformative technology in the realm of electronics and computing, particularly for neuromorphic systems. Their unique properties—such as non-volatility, scalability, and synapse-like behaviour—make them ideal for building efficient

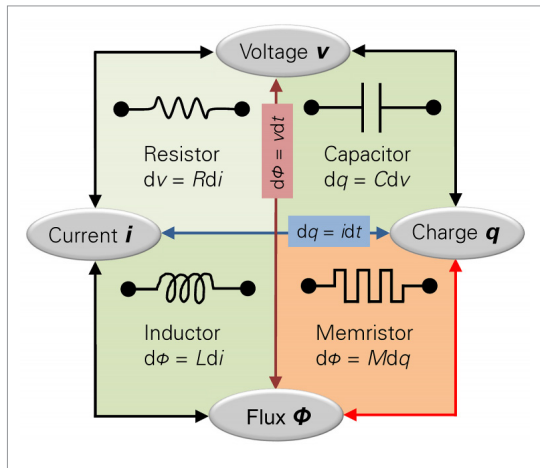


Fig. 2: Passive components to form memristor (Credit: <https://www.elprocus.com>)

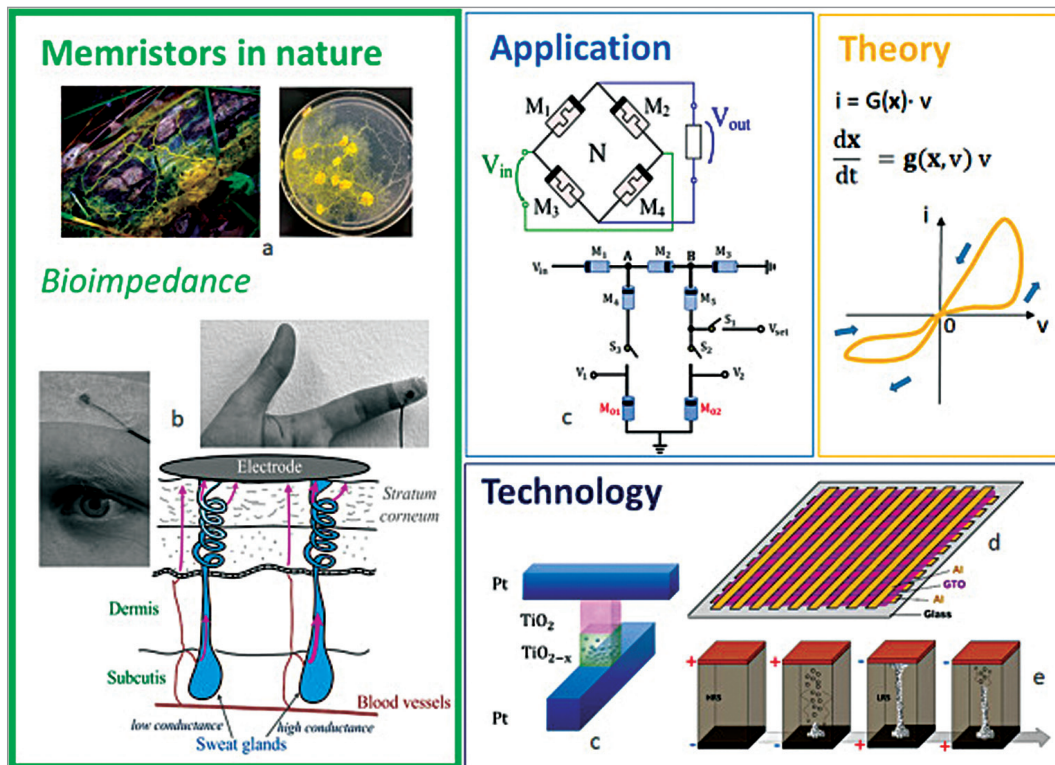


Fig. 3: Bioimpedance through memristors (Credit: <https://www.sciencedirect.com>)

and adaptive computing architectures. As research and development continue to advance, memristors are poised to play a pivotal role in the next generation of computing technologies, bringing us closer to the realisation of truly brain-like artificial intelligence. The journey from theoretical concept to practical application highlights the profound impact memristors will have on the future of electronics and neuromorphic computing. **EFY**